

The phase-dressed leading term in general dimension

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Abstract

Unit 202 closes the deterministic core of programme A. For continuous phase ξ and amplitude η on $\mathbb{R}^d$, exponents h, k with $\lambda = \min_i (h_i + 1)/(2k_i)$, minimiser set J , $m = |J|$, $p = 2\lambda$, and $s = Nu^k$,

$$\frac{\int_{(0,1]^d} \eta(u) u^h e^{-\beta s^2 + \beta s \xi(u)} du}{N^{-p} (\log N)^{m-1}} \longrightarrow 2^{m-1} \cdot 2 \cdot \frac{1}{(m-1)! \prod_{i \in J} 2k_i} \times \int_{(0,1]^d} \eta(\pi u) S_{p/2}^{(\beta)}(\xi(\pi u)) \prod_{i \notin J} u_i^{h_i - 2k_i \lambda} du$$

(`phase_leading_tendsto`, limit `phaseCoeff`), where π sets the minimising coordinates to zero and $S_{p/2}^{(\beta)}(a) = \int_0^\infty s^{p-1} e^{-\beta s^2 + \beta a s} ds$. The proof is the Taylor-in-phase argument of units 199–201 assembled with `tendsto_of_approx`: the $2K$ -term Taylor approximants converge by the shifted-exponent theorem; the N -side remainder is bounded by $M_\eta \tau_K \int u^h e^{-(\beta/4)N^2 u^{2k}}$, whose normalised version converges to a K -independent constant; and the distance of `phaseCoeff` to the finite coefficient sum is bounded by the same τ_K against the limiting face measure. Numerical check ($d = 2$, $h = (0, 1)$, $k = (1, 1)$, $\lambda = \frac{1}{2}$, $J = \{0\}$, $\xi(u) = u_1$, $\eta = 1$, $\beta = 1$): $N \cdot I_N = 1.2332, 1.2378$ at $N = 10^2, 10^3$ against $\int_0^1 S(v) dv = 1.2383$. Zero `sorry/axiom`; 9079 jobs.

1 The things formalised

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noncomputable def faceNorm (h k : Fin d → ℕ) (l : ℕ) : ℝ :=
  1 / (multCount (ratioExp h k) l - 1)! * ∑ i, if ratioExp h k i = l then 1/(2 * k i)
  else 1
noncomputable def phaseCoeff (h k) (l : ℕ) ( : (Fin d → ℕ) → ℕ) : ℝ :=
  2 ^ (multCount (ratioExp h k) l - 1) * 2 * faceNorm h k l *
  ∑ u in unitBox d, ( (faceProj h k l u) * phaseMoment (2 l) ( (faceProj h k l
    u)))
  * residualWeight h k l u
theorem sq_mul_prod_pow : (N * ∑ i, x i ^ k i) ^ 2 = N ^ 2 * ∑ i, x i ^ (2 * k i)
theorem quadKernel_eq_phase_mul_gauss : quadKernel (x) (N * x^k)
  = exp ((N * x^k) * ( * x)) * exp (-( * N^2 * x^(2k)))
theorem taylor_term_eq / phase_diff_eq (pointwise algebra of the truncated
  expansion)
theorem phase_remainder_le (hN : 0 ≤ N) (hb : x closedCube d, | * x| ≤ b)
  (hM0) (hM : x closedCube d, | x| ≤ M) (K) :
  | ∑ u^h quadKernel (u) (N u^k) - ∑ j < 2K, ^j/j! * (N^j * ( ^j )
    u^{h+jk} e^{-N^2 u^{2k}}) |
  ≤ M * tailCoeff b K * monomialBoxReal d h k (/4) (N^2)

```

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theorem phaseCoeff_remainder_le ... :
  |phaseCoeff h k l      - j < 2K, ^j/j! * (2^(m-1) * amplitudeCoeff
    (phaseShift h k j) k (1+j/2)  (^j ))|
    tailCoeff b K * (2^(m-1) * 2 * faceNorm h k l * (M * gaussTail (2l) *
      residualWeight))
theorem phase_leading_tendsto (hk) (hl) (h) (hmin) (hatt) (h : Continuous ) (h :
  Continuous ) :
  Tendsto (fun N => ( x in unitBox (d+1), x * (( x^h) * quadKernel ( x) (N *
    x^k)))
    / (N^(-(2l)) * log N ^ (multCount (ratioExp h k) l - 1))) atTop ( (phaseCoeff
    h k l ))

```

2 Mathematics

Three bookkeeping facts make the assembly exact. (i) $u^{h+jk} = u^h(u^k)^j$ and $N^j(u^k)^j = s^j$, so the j -th Taylor term of $e^{\beta s \xi}$ is literally the shifted-exponent integrand with amplitude $\xi^j \eta$ (`taylor_term_eq`). (ii) The Gaussian moments $\int_0^\infty s^{p-1} s^j e^{-\beta s^2} ds = \frac{1}{2} \Gamma(\lambda + j/2) \beta^{-\lambda-j/2}$ turn the finite coefficient sum $\sum_j \frac{\beta^j}{j!} 2^{m-1} \text{amplitudeCoeff}(h + jk, \dots)(\xi^j \eta)$ into $2^{m-1} \cdot 2 \cdot \text{faceNorm} \int \eta(\pi u) [\sum_j \frac{(\beta \xi(\pi u))^j}{j!} \frac{1}{2} \Gamma \beta^-] w(u) du$, the finite Taylor approximant of S under the face measure. (iii) Both remainders are controlled by the one tail constant τ_K of unit 199, uniformly because $|\beta \xi| \leq \beta \|\xi\|_\infty$ on the cube; the N -side normaliser $\int u^h e^{-(\beta/4) N^2 u^{2k}} / (N^{-2\lambda} (\log N)^{m-1})$ converges (bare monomial theorem at $\beta/4$, square-parameter transport) and is eventually $\leq L + 1$. Given ε , choose K with $\tau_K \max(C_1, C_2) \leq \varepsilon$ and apply `tendsto_of_approx`. No infinite series, no Cauchy limit, no $\Sigma/\int$ interchange beyond finite sums, as Astra #21 recommended.

3 Lean notes

- **Single-step calc followed by a dedented have:** after a one-step `calc a b := by ...`, the calc-steps parser saves its position at the *next token*, so a following `have h := e` at any column is parsed as a further calc step (a term-level `have`) and the parser dies at the next line with “expected :=” — here on the `at` of `rw [...] at h`. The symptom is a parse error two lines below the real culprit plus “unknown identifier” for everything the `have` defined. Fix: do not end a tactic block with a single-step calc (use `gcongr` directly), or give the calc two steps.
- `have h := ...` silently shadows the exponent vector `h`; the next `faceProj h k l` then fails with a `Prop-vs-Fin d → mismatch`. Name auxiliary facts `hnorm`.
- `gcongr` discharges the side goal $0 \leq M$ itself when it is an assumption; the remaining goal is only the main inequality $|x| \leq M$.
- Passing `_` for the continuous function in `integrableOn_face_mul ... _` (by `fun_prop`) leaves a metavariable that higher-order unification (`?f (faceProj u)`) cannot solve; give `f` explicitly.
- `sed 's/\bshiftExp\b/.../'` does not rename `prod_pow_shiftExp`: `_` is a word character. Unit 200’s lemmas are now `*_phaseShift` (renamed in this commit).

- `simp only [mul_sub, sub_mul, Finset.mul_sum, Finset.sum_mul]` closes the distributivity identity between a difference of products and a sum of products without `ring` (which cannot see inside `Finset.sum`).

4 Next

Unit 203: Headline XIII (paper kernel $e^{-\beta N^2 u^{2k} + \beta N u^k \xi}$, paper constant $1/((m-1)! \prod_{i \in J} k_i)$, equal-ratio corner form $\eta(0) S_{p/2}(\xi(0))/((d-1)! \prod k_i)$, zero-phase regression $\text{phaseCoeff}(\xi=0) = 2^{m-1} \text{amplitudeCoeff}$), mirror, review, report v2.